MILITARY SPECIFICATION

WIRE AND CABLE, ELECTRICAL, INSULATED, HIGH TEMPERATURE

1. SCOPE

1.1 Scope. - This specification covers irradiated modified polyolefin insulated hook up and general purpose wire and cable for use in temperature ranges as follows:

Temperature Range	Minimum Life
-55 degrees to 125 degrees centigrade	Continuous
-55 degrees to 150 degrees centigrade	1000 hours
-55 degrees to 175 degrees centigrade	96 hours
-55 degrees to 200 degrees centigrade	16 hours
-55 degrees to 225 degrees centigrade	4 hours
-55 degrees to 250 degrees centigrade	1 hour

1.2 Classification.

1.2.1 Hook up wire (unshielded). - Hook up wire shall be identified as follows:

TR	Class (see 1.2.3)
T	Conductor type (see 1.2.4)
22	Size (see 1.2.5)
(19)	Conductor (see 1.2.6)
U	Construction (see 1.2.7)
4	Color (see 1.2.9)

1.2.2 Cables. - Cables composed of one or more conductors shall be identified as follows:

TR	Class (see 1.2.3)
T	Conductor type (see 1.2.4)
22	Size (see 1.2.5)
(19)	Conductor (see 1.2.6)
S	Construction (see 1.2.7)
T	Shield type (see 1.2.4)
2	Number of conductors
R	Jacket (see 1.2.8)
1/3-9	Color (see 1.2.9)

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1.2.3 Class. - Hook-up wire and cable shall be furnished in the following classes:

Class TL for 300 volt service
Class TR for 600 volt service

1.2.4 Type. - Conductors and shields shall be of the following types: (see 3.4)

Type T tin coated copper
Type W tin coated copper clad steel
Type S silver coated copper

1.2.4.1 Flat shields. - When the individual strands of shielding material are composed of rectangular cross section, the letter "F" shall be added after the letter of paragraph 1.2.4 denoting type of shield.

1.2.5 Size. - The conductor size shall be indicated using American Wire Gage sizes.

1.2.6 Conductor. - The stranding of the conductor shall be shown in parentheses.

1.2.7 Construction. - The finished wire or cable construction shall be shown by the symbols "S" for shielded and "U" for unshielded construction.

1.2.8 Jacket. - Jackets over the shield and/or conductors will be as specified in 3.7.

1.2.9 Color code. - The color code shall consist of the symbols selected from Table I. A single symbol shall denote solid color, striped color codes shall consist of the symbol for insulation color followed by the symbol or symbols for each stripe color in the same sequence the stripe or stripes appear on the wire. Multiple conductor cables shall be designated with the slash (1) to distinguish conductor colors followed by the jacket color.

Table I

Symbol	Color	Symbol	Color
0	Black	5	Green
1	Brown	6	Blue
2	Red	7	Violet
3	Orange	8	Gray
4	Yellow	9	White

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

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SPECIFICATIONS

FEDERAL

J-C-98 - Cable and wire, insulated; Methods of sampling and testing.
O-C-141 - Carbon tetrachloride, technical grade
QQ-W-343 - Wire, electrical and non-electrical, copper, (Uninsulated).
QQ-W-345 - Wire, Electrical, Steel Copper - Covered.
QQ-S-571 - Solder Lead Alloy; Tin, Lead Alloy; Tin Alloy; Flux Cord, Ribbon and Wire and Solid Form.
TT-N-95 - Naptha Aliphatic.
VV-I-530 - Insulating Oil, Electrical (For Transformers, Switches, and Circuit Breakers).

MILITARY

MIL-A-6091 - Alcohol, Ethyl, Specially Denatured Aircraft.
MIL-C-915 - Cable, Cord and Wire, Electrical (Shipboard Use).
MIL-C-45662 - Calibration of Standards.
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-F-5566 - Fluid, Anti-Icing (Isopropyl Alcohol).
MIL-G-5572 - Gasoline, Aviation; Grades 80/87, 91/96, 100/130, 115/145.
MIL-H-5606 - Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance.
MIL-H-7083 - Hrdraulic Fluid, Non-Flammable, Aircraft, Hydrolube.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance.
MIL-J-5624 - Jet Fuel; Grades JP-3, JP-4 and JP-5.
MIL-L-6082 - Lubricating Oil; Aircraft Reciprocating Engine (Piston).
MIL-L-7808 - Lubricating Oil; Aircraft Turbine Engine, Synthetic Base.
MIL-O-6081 - Oil; Lubricating Jet Engine.
MIL-P-7254 - Propellant, Nitric Acid.
MIL-P-10025 - Packaging, Packing, and Marking for Interplant Shipment of Inert Ammunition Components, General Specification for.
MIL-P-26536 - Propellant, Hydrazine.
MIL-P-26539 - Propellant, Nitrogen Tetroxide.
MIL-P-27402 - Propellant, Hydrazine -UNS- Dimethyl Hydrazine (50% N_2H_4-50% UDMH).
MIL-T-5438 - Tester, Abrasion, Electrical Cable.
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature.

COMMERCIAL

Skydrol 500A - Hydraulic Fluid (Monsanto Chemical Co.).

STANDARDS

MILITARY

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MIL-STD-104 - Limits for Electrical Insulation Color.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

AEROSPACE INDUSTRIES ASSOCIATION OF AMERICA INCORPORATED

NAS-703 - Wire, Electrical, Insulated, Copper, High Temperature, Hook Up and General Purpose (For 200 degree Centigrade Service).

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. - The wire and cable furnished under this specification shall be a product which has been tested and has passed the initial production tests specified herein:

3.2 Selection of materials. - All materials used in the construction of the completed wire and cable shall be of the quality and form best suited for the purpose intended. Where materials, methods of test, or other requirements are not specifically designated, it is understood that the best commercial practice shall apply.

3.3 Construction.

3.3.1 Hook-Up wire. - Hook-up wire shall consist of the following:

a. Conductor as specified (see 1.2).
b. Primary insulation (see 3.5).

3.3.2 Cables. - Cables shall consist of the following:

a. One or more hook-up wires (see 3.3.1).
b. A shield composed of woven braid (see 3.6), if designated (see 1.2.2).
c. An outer jacket (see 3.7), if designated (see 1.2.2).

3.4 Conductors. - Unless otherwise specified, the inner conductor(s) of the wire or cables shall be in accordance with Table II. The materials for the conductor or conductors shall be as specified in this section inclusive, as applicable, and shall meet the requirements of this section prior to being incorporated into finished hook-up wire or cable.

3.4.1 Copper wire. - Unless otherwise specified, copper wire shall conform to uncoated or coated Type S wire of QQ-W-343, when tested as specified in 4.4.1.1.

3.4.2 Copper covered steel wire. - Unless otherwise specified, coated copper covered steel wire shall conform to Type I of QQ-W-345, when tested as specified in 4.4.1.1.

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3.4.3 Tinned solid wire and strands. - Where specified, copper and copper covered steel wire shall be coated with a continuous uniform covering of commercially pure tin in accordance with QQ-W-343 when tested as specified in 4.4.1.1.

3.4.3.1 Silver coated stranded wire. - When specified, wire shall be coated with a continuous uniform covering of silver, not less than 40 micro inches in thickness. The coating shall be in accordance with MIL-W-16878 when tested as specified in 4.4.1.1.

3.4.4 Stranding. - Stranding shall be concentric bunch, concentric strand or rope-lay, in accordance with Table II, unless otherwise specified. Other acceptable strandings are those shown in QQ-W-343. The direction of lay of the outside layer of stranded conductors shall be left hand and the length of lay shall be in accordance with Table II. For rope-lay stranded conductors, the length of lay of the wires composing the bunch stranded members shall not be more than 30 times the diameter of the members, and the length of lay of the outer layer of rope-lay stranded conductor shall not be less than 8 nor more than 16 times the outside diameter of the completed conductor. Check the physical measurements specified in 4.3.2.1.1 test resistance as specified in 4.4.1.2.

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TABLE II

Conductor Size (AWG)	Number of Strands	Strand Size (AWG)	Max. Conductor Diameter (Inches)	Nominal Conductor (Cir. Mils)	Max. Conductor Resistance (Ohms/ 1000' at 20°C)	Length of Lay (Inches Max.)
30	7	38	.013	112	104	.25
28	7	36	.016	175	72.5	.25
26	7	34	.020	278	40.1	.25
24	19	36	.027	475	23.5	.33
22	19	34	.033	754	14.8	.33
20	19	32	.041	1216	9.16	.50
18	19	30	.052	1900	5.86	.80
16	19	29	.061	2432	4.58	1.0
14	19	27	.076	3838	2.90	1.2
12	19	25	.096	6080	1.83	1.5
10	37	26	.128	9361	1.19	1.8
8	133	29	.176	17,024	0.667	(para. 3.4.4)
6	133	27	.217	26,866	0.423	(para. 3.4.4)
4	133	25	.270	42,560	0.267	(para. 3.4.4)
2	665	30	.315	66,500	0.174	(para. 3.4.4)
0	1064	30	.410	106,400	0.111	(para. 3.4.4)

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3.4.5 Splices.

3.4.5.1 Solid conductor. Each conductor shall be continuous in length on any reel shipped. Solid conductor may be spliced, in which case they shall be electro-welded or brazed with silver (Ag) composition solder conforming to QQ-S-571. Splices shall be of the butt type except that solid conductors of AWG size 28 and smaller may be twisted, in which case they shall have a minimum overlap of 1/2 inch. Splices shall be so constructed and distributed throughout the conductor that the diameter, configuration, direct current resistance, flexibility, and mechanical strength of the completed conductor are not adversely affected (see 3.8, 3.10, 3.13 and 3.19).

3.4.5.2 Stranded conductors. Each conductor shall be continuous in length on any reel shipped. Splices shall not be made in a stranded conductor as a whole. However, individual strands of stranded conductors may be spliced, in which case they shall be electro-welded or brazed with silver (Ag) composition solder conforming to QQ-S-571. Splices shall be of butt type except that splices in strands of AWG size 28 and smaller may be twisted, in which case they shall have a minimum overlap of 1/2 inch. Splices shall be so constructed and distributed throughout the conductor that the diameter, configuration, direct current resistance, flexibility, and mechanical strength of the completed conductor are not adversely affected. (see 3.4.4, 3.10 and 3.19).

3.4.6 Solderability (wettability). The finished conductor, after being stripped to comply with 3.5.3 shall be tested for solderability as specified in 4.4.1.3 and comply with MIL-STD-202, method 208.

3.5 Primary insulation. The primary insulation shall be an irradiated, thermally-stabilized, flame-resistant, modified polyolefin composition. The insulation shall be concentrically applied so that at any cross-section along the length of the completed wire the concentricity shall be a minimum of 70 percent when measured in accordance with 4.4.2.1.

3.5.1 Physical properties. Physical properties of the primary insulation of the wire shall be as specified in Table III and measured in accordance with 4.4.2.2.

Table III

Characteristic	Minimum required value
Tensile Strength	2500 psi
Elongation	100%

Note: To maintain flexibility, where the primary insulation wall is greater than 0.015, the tensile strength shall not be less than 2000 psi.

3.5.2 Insulation flaws. One hundred percent of the wire, after application of the insulation, shall be subjected to the following potentials without failure in accordance with 4.4.2.3.

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Class TL - 2500 volts rms
Class TR - 4000 volts rms

3.5.3 Strippability. - The insulation of the finished conductor shall be capable of being readily stripped from the conductor by any standard thermal stripping technique leaving the conductor clean for soldering when tested as specified in 4.4.2.4.

3.6 Shield. - The shield shall be a closely woven braid of the type designated. The individual strands shall conform to 3.4. The braid shall have good "push-back" characteristics and provide not less than 90 percent coverage, determined as specified in 4.4.3.

3.7 Jacket. - The outer jacket shall be of similar composition to the primary insulation. It shall be applied concentrically, as in 3.5, and with a thickness as in Table IV. The jacket material shall meet the requirements of 3.5 hereof except that the voltage of 3.5.2 shall be 2000 volts rms, and the tensile strength of 3.5.1 shall be 2000 psi.

Table IV

Diameter over Shield (inches)	Diameter Increase (inches)
.031 to .059	.014 to .024
.060 to .119	.016 to .028
.120 and over	16% to 24% of the shielded diameter to a maximum increase of .200

3.8 Dimensions. - The dimensional requirements for hook-up wires, Class TL and Class TR, unshielded, are shown in Table V. (see 4.3.2.1.2).

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Table V

Class TL			Class TR		
Conductor Size (AWG)	Hook-up Wire Min. OD (inches)	Hook-up Wire Max. OD (inches)	Conductor Size (AWG)	Hook-up Wire Min. OD (inches)	Hook-up Wire Max. OD (inches)
			30	0.027	0.033
28	.024	.030	28	.029	.035
26	.029	.035	26	.034	.040
24	.034	.040	24	.040	.047
22	.041	.047	22	.046	.054
20	.049	.055	20	.054	.062
18	.060	.066	18	.065	.073
16	.070	.078	16	.078	.086
			14	.096	.104
			12	.118	.126
			10	.146	.160
			8	.205	.219
			6	.267	.281
			4	.320	.340
			2	.392	.416
			0	.500	.524

3.9 Heat resistance. - Heat resistance tests are applicable to hook-up wire and finished cable. After subjection to the heat resistance tests specified in 4.4.5, the specimens shall exhibit no cracks and shall be capable of withstanding without breakdown for one minute the application of voltage when applied in accordance with 3.19. Shrinkage of the primary insulation shall not exceed 1/8 inch.

3.10 Cold bend. - Cold bend tests are applicable to hook-up wire and finished cable. The specimens shall withstand a temperature of -55°C without cracking when bent over a mandrel as specified in 4.4.6. After bending, the wire shall be capable of withstanding without breakdown for one minute the application of voltage when applied in accordance with 3.19.

3.11 Water and solvent resistance. - After being subjected to the immersion test of 4.4.7, the finished wire shall show no evidence of cracking and shall be capable of withstanding without breakdown for one minute the application of voltage when applied in accordance with 3.19. Voltage shall then be increased at a uniform rate of approximately 500 volts per second until failure occurs. The breakdown voltage shall be in excess of 2500 volts rms for Class TL wire and in excess of 4000 volts rms for Class TR wire. The increase in diameter of the finished wire shall not exceed 5 percent.

3.12 Flammability. - When tested in accordance with 4.4.8, the burning time of the finished wire, and cables which have been round braid shielded and jacketed, shall not exceed 30 seconds and the flame travel upwards shall not exceed 3 inches.

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3.13 Resistance. - The resistance of the finished wire shall be tested in accordance with 4.4.9. For each type conductor the maximum allowable resistance shall be as shown in Table II multiplied by a factor as follows:

Type	Factor
T	1.08
W	2.70
S	1.00

3.14 Insulation resistance. - The finished wire shall have a minimum insulation resistance of 5000 megohms per 1000 feet when tested in accordance with 4.4.9. Cable jackets shall have a minimum insulation resistance of 30 megohms per 1000 feet when tested in accordance with 4.4.10.

3.15 Color. - The color of the finished wire, striping thereon, and the color of cable jackets shall conform to MIL-STD-104.

3.15.1 Striping (see 4.3.2.1.02). - The striping shall be spirally applied with a lay not exceeding 18 times the outside diameter of the primary insulation. The width of the first stripe shall not be less than 0.040 inch. Where two or three stripes are employed the width of the narrow stripe shall not be less than 0.015 inch. The distance between successive stripes shall not be less than two-thirds the width of the individual stripes.

3.16 Fungus resistance. - The finished wire shall be fungus inert when tested in accordance with MIL-E-5272.

3.17 Corrosion. - Specimens of hook-up wire and cable which have been subjected to the corrosion test of 4.4.11 shall be examined for attack of the conductor, shield (where applied) and test conductors. There shall be no evidence of pitting or corrosion of the outside coating or plating. In addition, there shall be no evidence of exposed copper. Darkening of the tin coating caused by normal air oxidation shall not be cause for rejection.

3.18 Soldering. - The insulating material shall not flare away from the conductor nor shall it shrink back along the conductor more than 1/16 inch when tested in accordance with 4.4.12.

3.19 Dielectric strength. - The finished wire and cable shall be capable of withstanding, without breakdown, the application of the following potentials when tested as specified in 4.4.13.

Class TL - 1600 volts rms
Class TR - 2500 volts rms

3.20 Radiation quality assurance.

3.20.1 Continuous monitoring. - The manufacturer shall provide automatic and continuous monitoring of all wire and cable produced in accordance with this specification.

3.20.1.1 Radiation by high voltage electrons.- Where irradiation is accomplished by use of high-voltage electrons, the current derived by capture of electrons in the metallic conductor en route to ground shall be measured by a suitable ammeter to indicate the continuing exposure of the wire or cable to the electron stream.

3.20.1.2 Radiation by other than high voltage electrons. - Where irradiation of wire and cable is accomplished by radiation other than high-voltage electrons, appropriate instrumentation will be provided to comply with the intent of 3.20.1.1.

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3.21 Radiation resistance. - After being subjected to the radiation resistance test of 4.4.14 specimens shall pass the dielectric strength requirement of 3.19.

3.22 Surface resistance. - The surface resistance shall not be less than 5 megohms when measured in accordance with 4.4.15.

3.23 Abrasion resistance. - The finished hook-up wire shall meet the minimum requirements specified in Table XI. The method of testing shall be in accordance with 4.4.16.

3.24 First article testing. - Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.25 Workmanship. - The workmanship shall be first class in every respect and uniform throughout each lot of wire or cable offered for delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. - When the contract requires first article inspection, the contractor shall submit to the testing facility designated in the invitation for bids or request for proposal 150 feet of wire and any designated cable construction of sizes 20, 16 and 8 AWG. When approved each size will qualify additional sizes as follows:

20 AWG will qualify 30 thru 20 AWG
16 AWG will qualify 18 thru 14 AWG
8 AWG will qualify 12 thru 0 AWG

4.2.2 Inspection and testing to be performed. - The first article inspection sample shall be subjected to the inspections and tests specified in Table VI. Tests need not be conducted in order specified in Table VI.

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Table VI

First article inspections and tests

Examination or test	Contractor's inspection permissible before submission	Destructive	Non-Destructive	Wire	Sub-Assemb.	Assemb.
1. Classification of characteristics (see 4.3.2.1)	X		X	X	X	X
2. Construction (see 3.3)	X		X	X	X	X
3. Conductors (see 3.4)		X		X		
4. Primary Insulation (see 3.5)		X		X		
5. Shield (see 3.6)		X			X	
6. Jacket (see 3.7)		X				X
7. Heat Resistance (see 3.9)		X		X		X
8. Cold bend (see 3.10)		X		X		X
9. Water & solvent resistance (see 3.11)		X		X		
10. Flammability (see 3.12)		X		X		X
11. Resistance (see 3.13)	X		X	X		
12. Insulation resistance (see 3.14)	X		X	X		X
13. Color (see 3.15)	X		X	X	X	X
14. Fungus resistance (see 3.16)		X		X		

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Examination or test	Contractor's inspection permissible before submission	Destructive	Non-Destructive	Wire	Sub-Assemb	Assemb
15. Corrosion (see 3.17)		X		X		X
16. Soldering (see 3.18)	X		X	X		
17. Dielectric strength (see 3.19)		X		X		X
18. Radiation quality assurance (see 3.20)	X		X	X	X	X
19. Radiation resistance (see 3.21)		X		X		X
20. Surface resistance (see 3.22)		X		X		X
21. Abrasion resistance (see 3.23)		X		X		

4.2.3 Inspection equipment. - The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.

4.2.4 Rejection. - If any test or measurement fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. - The formation of lots and related functions shall be in accordance with sections 5 and 6 of MIL-STD-105. All lots shall consist of units not greater than 200,000 feet each. A lot shall consist of all completed wire and cable made on any single machine at any one time. Each spool shall show the lot number.

4.3.1.1 Inspection lot. - An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. - Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

4.3.2.1 Classification of characteristics. - Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

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4.3.2.1.1 Conductor stranding (see 3.4.4).

Categories	Characteristics	Method of inspection
Critical: None defined		
Major:		
101.	Conductor diameter, max	Gage
102.	Length of lay, max	Gage
103.	Not more than one length on reel (3.4.5.2)	Visual-Manual
Minor:		
201.	Workmanship (see 3.25)	Visual-Manual

4.3.2.1.2 Hook-up wire Class TL and TR (see 3.8).

Categories	Characteristics	Method of inspection
Critical: None defined		
Major:		
101.	Diameter of conductor (see Table V)	Gage
102.	Not more than one length on reel not spliced (see 3.4.5.2)	Visual-Manual
Minor:		
201.	Workmanship (see 3.25)	Visual-Manual

4.3.2.1.3 Striping of finished wire (see 3.15.1).

Categories	Characteristics	Method of inspection
Critical: None defined		
Major:		
101.	Applied lay does not exceed 18 times the outside diameter of the primary insulation	Gage
102.	Width of first stripe more than 0.040 inch	Gage
103.	Where more than one stripe is employed, the width of the second stripe is more than 0.015 inch width	Gage
104.	Distance between successive stripes is more than 2/3 the width of individual stripes	Gage
Minor:		
201.	Workmanship (see 3.25)	Visual-Manual

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4.3.3 Inspection testing. All the tests specified in Table VII shall be performed.

Table VII

	Test	Section	Classification
a.	Conductor strands	3.4	Major
b.	Stranded conductors	3.4.4	Major
c.	Primary insulation	3.5	Major
d.	Shield coverage	3.6	Major
e.	Jacket	3.7	Major
f.	Heat resistance	3.9	Major
g.	Cold bend	3.10	Major
h.	Resistance	3.13	Major
i.	Insulation resistance	3.14	Major
j.	Color	3.15	Major
k.	Soldering	3.18	Minor
l.	Dielectric strength	3.19	Major

Of these tests a, b, c, d, e, f, g, k and l are considered destructive and samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of wire or cable required by the contract.

4.3.3.1 Conductors.

4.3.3.1.1 Conductor strands. Each conductor prior to bunching or stranding shall be tested in accordance with 4.4.1.1. If any conductor fails to comply with 3.4.1, 3.4.2, 3.4.3 and 3.4.3.1 as applicable, the lot of wire or cable shall be rejected.

4.3.3.1.2 Stranded conductors. Each stranded conductor shall be tested as specified in 4.4.1.2. If the resistance fails to comply with Table II (see 3.4.4), or QQ-W-343 when authorized, the conductor shall be rejected.

4.3.3.1.3 Solderability (wettability). Each lot of finished conductors shall be tested as specified in 4.4.1.3. If any sample fails to comply with the requirements of 3.4.5, the lot shall be rejected.

4.3.3.2 Primary insulation.

4.3.3.2.1 Concentricity. The primary insulation of the wire after application of the insulation shall be tested in accordance with 4.4.2.1. If the primary insulation fails to comply with 3.5, the lot of wire shall be rejected.

4.3.3.2.2 Physical properties. The primary insulation of the wire after application of the insulation shall be tested in accordance with 4.4.2.2. If the primary insulation fails to comply with 3.5.1, the lot of wire shall be rejected.

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4.3.3.2.3 Insulation flaws. One hundred percent of the wire after application of the insulation shall be tested in accordance with 4.4.2.3. If any wire fails to comply with 3.5.2, the insulated wire shall be rejected.

4.3.3.2.4 Strippability. Each lot of insulated conductors shall be tested as specified in 4.4.2.4. If any sample fails to comply with the 3.5.3, the lot shall be rejected.

4.3.3.3 Shield coverage. The shield coverage shall be calculated as specified in 4.4.3. If any shield fails to comply with 3.6, the wire or cable shall be rejected.

4.3.3.4 Jacket. The jacket of cable or wire shall be tested as specified in 4.4.4. If any jacket fails to comply with 3.7, the wire or cable shall be rejected.

4.3.3.5 Heat resistance. Specimens of each wire or cable shall be tested in accordance with 4.4.5.1 If any specimen fails to comply with 3.9, the wire or cable shall be rejected.

4.3.3.6 Cold bend. The finished wire shall be tested in accordance with 4.4.6. If any sample fails to comply with 3.10, the wire shall be rejected.

4.3.3.7 Resistance. Samples of finished wire shall be tested in accordance with 4.4.9. If any sample fails to comply with 3.13, the wire shall be rejected.

4.3.3.8 Insulation resistance. Samples of wire and cable shall be tested in accordance with 4.4.10. If any sample fails to comply with 3.14, the wire and cable shall be rejected.

4.3.3.9 Color. The color of finished wire and striping thereon shall be tested in accordance with MIL-STD-104. If the color fails to comply with MIL-STD-104, the wire shall be rejected.

4.3.3.10 Soldering. Specimens shall be tested in accordance with 4.4.12. If any specimen fails to comply with 3.18, the wire or cable shall be rejected.

4.3.3.11 Dielectric strength. Specimens of wire or cable shall be tested in accordance with 4.4.13, as applicable. If any specimen fails to comply with 3.19, the wire or cable shall be rejected.

4.3.4 Inspection equipment. The contractor shall submit for approval to Commanding Officer, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-NR4, a complete list of inspection equipment including instructions for use and drawings required for special equipment for each inspection required in this specification prior to fabricating the item.

4.4 Test methods and procedures. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Conductors.

4.4.1.1 Conductor strands (see 3.4.1, 3.4.2 and 3.4.3). - Copper wire shall be tested as specified in QQ-W-343. Copper covered steel wire conductors shall be tested as specified in QQ-W-345. Tinned solid wire conductors strands shall be tested as specified in QQ-W-343. Silver coated stranded wire conductors shall be tested as specified in MIL-W-16878.

4.4.1.2 Stranded conductors (see 3.4.4). - Measure the resistance of the conductor with an approved resistance measuring device.

4.4.1.3 Solderability (wettability) (see 3.4.6). - A 12 inch sample shall be taken from each lot of finished conductors, stripped as specified in 4.4.2.4, and prepared and tested as specified in MIL-STD-202, Method 208. After completion of the test the conductor shall be tested as specified in MIL-STD-202, Method 208.

4.4.2 Primary insulation.

4.4.2.1 Concentricity (see 3.5). - The concentricity of the primary insulation shall be determined by dividing the minimum wall thickness by the maximum wall thickness. The minimum and maximum wall thicknesses shall be measured under suitable magnification from the outermost rim of the individual strand to the outer edge of the insulation.

4.4.2.2 Physical properties (see 3.5.1). - The determination of tensile strength and elongation shall be made in accordance with J-C-98, Methods 3021 and 3031, respectively, utilizing 1 inch bench marks and 1 inch jaw separation.

4.4.2.3 Insulation flaws (see 3.5.2). - The insulation flaws test shall be conducted in accordance with J-C-98, Method 6211.

4.4.2.4 Strippability (see 3.5.3). - A 12 inch sample shall be taken from each lot and stripped 1 1/8 inch minimum from the end using any standard thermal stripping technique. The stripped portion of the sample shall be examined with any optical device capable of 10 power magnification to assure that no evidence of the insulation remains on the conductor.

4.4.3 Shield coverage (see 3.6). - Shield coverage shall be calculated using the following formula:

$$\text{Percent coverage} = (2F - F^2)\ 100;$$

$$F = \frac{NPd_1}{\text{Sin } a};\quad \tan a = \frac{2\pi(D + 2d_2)\ P}{C}$$

where:

N = Number of strands per carrier

P = picks per inch

d_1 = diameter of strand (inches); width if flat braid used

d_2 = diameter of strand (inches); thickness if flat braid used

a = angle of shield with cable axis (degrees)

D = diameter of cable under shield (inches)

C = number of carriers

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4.4.4 Jacket (see 3.7). - Measure the diameter over the shield. Measure the diameter over the jacket. Subtract the diameter of the shield from the diameter of the jacket. The jacket shall also be tested in accordance with 4.4.2 using test criteria in 3.7.

4.4.5 Heat resistance (see 3.9).

4.4.5.1 Hook-up wire and cable. - One inch of the insulating compound shall be removed from each end of a twenty-inch sample of the finished wire or from each wire of a finished cable. The central portion of the specimen shall then be bent half-way around a coated stainless steel mandrel having a diameter as specified in Table VIII. The ends of each conductor shall be tied together and then loaded as specified in Table VIII in such a manner that the portion of the insulation between the conductor and the mandrel is under compression while the conductor is under the tension specified in Table VIII. This condition shall be maintained for a period of four hours in an air-circulating type oven, maintained at the temperature of 225 ± 4°C. The oven shall be equipped with a temperature-recording chart whose thermocouple shall be centrally located in the oven at the same height as the mandrel and not more than two inches from the specimen. The velocity of air past the specimen shall not exceed 50 feet per minute. After completion of the high temperature conditioning, the specimen shall remain in the oven and be cooled to between 20°C and 25°C within a period of one hour. When cooled, the specimen shall be removed from the oven, freed from tension, removed from the mandrel, and straightened by securing one end of the specimen to the mandrel and the other end to the load-weight specified in Table VIII. The mandrel shall be rotated at a rate of 12 ± 1 revolutions per minute until the full length of the specimen is wrapped around the mandrel and is under the specified tension with adjoining coils in contact. The mandrel shall then be rotated at the same rate in the reverse direction until the full length of the specimen which was outside during the first wrapping is now next to the mandrel. This procedure shall be repeated until two bends in each direction have been formed in the same section of the specimen. The specimen shall then be subjected to the dielectric strength test of 4.4.13 as applicable.

Table VIII

Wire Size (AWG)	Mandrel Diameter (Inches)	Loading (Pounds)	
		Class TL	Class TR
30 - 24	1/4	0.125	0.25
22 - 20	3/8	0.375	0.75
18 - 16	1/2	0.500	1.0
14 - 12	3/4		2.0
10 - 8	1		3.0
6 - 4	2		4.0
2 - 0	3		5.0
Cable	20 x Diameter	Individual Wires loaded as above	

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4.4.6 Cold bend (see 3.10). - One end of a previously untested specimen of wire and cable, of suitable length, shall be secured to the applicable mandrel specified in Table IX, in such a manner that bending will occur at a distance at least six inches from one end of the specimen. At least two turns shall be made about the mandrel at room temperature. The load weight specified in Table IX shall be attached to the other end of hook-up wire specimens and twice the weight given in Table IX shall be attached to the free end of each individual wire in a cable specimen. The temperature of the specimen shall be lowered to -55°C. After this temperature has been maintained for four hours, the specimen, while still at this low temperature, shall be wrapped two turns around the mandrel in a period of 1/2 minute, by revolving the mandrel. After removal from the low temperature cabinet, the specimen shall be returned to room temperature and subjected to the dielectric test strength test of 4.4.13 as applicable.

Table IX

Wire Size (AWG)	Mandrel Diameter (Inches)	Loading (Pounds)
30 - 26	3/8	0.5
24 - 22	1/2	0.75
20 - 16	1	1.0
14 - 12	1-3/4	3.0
10 - 8	2-1/4	5.0
6 - 4	4	15.0
2 - 0	6	25.0
Cable	20 x diameter	see above

4.4.7 Water and solvent resistance (see 3.11). The diameters of separate specimens of wire at least 24 inches long shall be measured and recorded and then the specimens shall be immersed to within six inches of their ends in the fluids indicated in Table X for the times and temperatures indicated. Upon removal, the test specimens shall be carefully wiped dry and the diameters shall again be measured and recorded. The specimens shall then be allowed to remain at room temperature for one hour. At the end of this period, each specimen shall be subjected to the dielectric strength test of 4.4.13 as applicable.

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Table X

Fluid	Specification	Hours	Temperature, °C
a) Carbon Tetrachloride	O-C-141	12	20-25
b) Insulating Oil	VV-I-530	20	20-25
c) Naptha	TT-N-95	20	20-25
d) Anti-Icing Fluid	MIL-F-5566	20	20-25
e) Aviation Gasoline, Grade 100/130	MIL-G-5572	see note	20-25
f) Hydraulic Fluid	MIL-H-5606	20	20-25
g) Jet Fuel, Grades JP-4, JP-5	MIL-J-5624	20	20-25
h) Jet Engine Oil	MIL-O-6081	20	20-25
i) Lubricating Oil	MIL-L-6082	20	20-25
j) Ethyl Alcohol	MIL-A-6091	20	20-25
k) Hydraulic Fluid	MIL-H-7083	20	20-25
l) Nitric Acid	MIL-P-7254	1/2	20-25
m) Lubricating Oil	MIL-L-7808	see note	20-25
n) Salt Water (5% salt)	-----	20	20-25
o) Skydrol 500A	Monsanto	48	20-25
p) Hydrazine	MIL-P-26536	1/2	20-25
q) Nitrogen Tetroxide	MIL-P-26539	1/2	20-25
r) 50% Hydrazine - 50% UDMH	MIL-P-27402	1/2	20-25

Note: A specimen shall be immersed in a 50-50 mixture of (e) and (m) for 20 hours.

4.4.8 Flammability (see 3.12). The test specimen shall be 24 inches in length, marked at a distance of 6 inches from the lower end to indicate the central point for flame application. The specimen shall be placed at an angle of 45 degrees within a chamber approximately 2 by 1 by 1 foot, open at the top and one vertical side (front), and which allows sufficient flow of air for complete combustion, but which is free from draft. The specimen shall be parallel to and approximately 6 inches from the front of the chamber. The upper end of the specimen shall pass over a pulley and shall have the appropriate weight specified in Table VIII attached to it so that the specimen is held taut throughout the flammability test. A flame from a Bunsen burner shall be applied for 30 seconds at the lower test mark. The Bunsen shall have a 1/4 inch inlet, a nominal bore of 3/8 inch, and a length of approximately 4 inches from top to primary inlets. The burner shall be adjusted to produce a 2-inch flame with an inner cone one-third of the flame height. The burner height shall be adjusted so that the hottest portion of the flame is applied to the test mark on the wire. The time of burning and distance of flame travel upward along the wire from the test mark shall not exceed the values specified in 3.12. (Note: Due to conductor breakage, this test is not applicable to conductors of AWG 24 or smaller).

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4.4.9 Resistance (see 3.13). The finished wire shall be tested for resistance in accordance with J-C-93, Method 6021. Multiply the resistance in Table II for each type of conductor by the factor in 3.13.

4.4.10 Insulation resistance (see 3.14). The insulation resistance shall be measured as soon as practicable after the dielectric resistance test od 3.19. The insulation resistance test shall be in accordance with section 4.8.25 of MIL-C-915.

4.4.11 Corrosion (see 3.17). A specimen six inches in length, of previously untested single conductor hook-up wire or jacketed cable shall be cleaned by wiping with a soft clean cloth moistened with acetone CP and allowed to dry two hours at room temperature. A six-inch length of tin coated 22 AWG copper conductor, stranded 19 x 34 AWG shall be wrapped around the cleaned specimen with a uniform tension applied as follows: Fix the weights indicated in Table XI to the free end of the wire and rotate the sample so that the wire will be wound on the sample with uniform tension.

Table XI

Outside Diameter (Inches)	Weight (Pounds)
.064 and smaller	0.1
.065 to .204	0.2
.205 and larger	0.3

The prepared specimen shall be hung in an air gravity convection type oven maintained at a temperature of 135° ± 3°C for a period of 7 days. It shall be removed from the oven at the end of this period and conditioned at room temperature for one hour. The externally wrapped conductor shall be carefully removed. The insulation of the single conductor hook-up wire and the jacket on the cable shall be very carefully slit open with a razor blade and peeled back manually to expose the inner conductor or shield. The outer wrapped test conductor, inner conductor, and shield shall then be carefully examined for signs of pitting, corrosion or removal of the tin coating sufficient to expose bare copper.

4.4.12 Soldering (see 3.18). Specimens of insulated wire six inches in length shall be prpeared for test by removing 1/2 inch of insulation at one end. At a point 1/2 inch from the skinned end of the wire, the specimens shall be given a 90° bend over mandrels of their own diameters. These ends shall be then immersed for five seconds, to within 1/8 inch of the insulation, in a pot of molten 60-40 (tin-lead) solder maintained at a temperature of approximately 320°C. The distance the insulation has shrunk back along the conductor shall then be measured.

4.4.13 Dielectric strength (see 3.19).

4.4.13.1 Specimens. Unless otherwise specified (see 4.4.13.1.1), the dielectric strength test shall be made on 100 percent of the finished wire and cable.

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4.4.13.1.1 Specimens for missile application. Before testing for dielectric strength, a 5 foot sample of finished wire or cable shall be selected at random from the lot and subjected to a vacuum of $.256 \times 10^{-4}$ pounds per square inch, min. (altitude of 3000,000 ft.) for a period of 10 minutes minimum. After completion of the test the specimen shall be tested as specified in 4.4.13.2.

4.4.13.2 Procedure.

4.4.13.2.1 Class TR hook-up wire. The uninsulated ends of the specimens shall be fastened in metallic contact to a metal bar. These specimens shall be immersed to within 6 inches of their ends in a water bath consisting of 0.5 percent wetting agent such as Aerosol OT. The water bath shall be maintained at a normal room temperature. The specimens shall remain immersed for 2 hours prior to the application of the potential. With the specimens still immersed, the potential specified in 3.19 shall be applied between the conductor and water bath, which serves as the ground electrode for one minute. The initially applied voltage shall not be greater than 600 volts, and the rate of increase shall be 500 volts/second.

4.4.13.2.2 Class TL hook-up wire. The dielectric strength of Class TL hook-up wire shall be determined in accordance with J-C-98, Method 6211, using water as the electrode.

4.4.13.2.3 Cable. The test procedure for cables shall be identical with 4.4.13.2.1 except that cables shall not be immersed. The voltage shall be applied between conductors and between the shield and each conductor in turn.

4.4.14 Radiation resistance (see 3.21). A sample of hook-up wire or unshielded conductor to be incorporated into cable shall be subjected to a radiation dosage of 500 megarads at an average rate of between 5 and 10 megarads per minute. Following exposure, the center portion of the sample shall be wound for at least ten turns over a mandrel whose diameter is ten times the overall diameter of the wire or cable. The ends of the sample shall extend at least six inches beyond the wound portion. The specimen shall be moved without unwinding and shall be subjected to the dielectric strength test of 4.4.13 as applicable.

4.4.15 Surface resistance (see 3.22). The wire shall be tested in accordance with Section 4.4.3.2 of MIL-W-16878.

4.4.16 Abrasion reistance (see 3.23). The wire shall be tested in accordance with the test method of Section 4.3.11 of NAS 703 and Table XII.

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Table XII

Wire Size	Class	Weight Support Bracket	Weight to be Added (lbs.)	Tape Travel (Inches)	Wire Tension (lbs.)
AWG 20	TL	A	.25	15	1
	TR	A	.50	20	1
AWG 16	TL	B	.50	20	1
	TR	B	.75	20	2
AWG 8	TR	B	1.50	90	2

5. PREPARATION FOR DELIVERY

5.1 Packing. The wire and/or cable shall be packed in accordance with the minimum package and pack which will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity. Shipping containers shall be snugly packed and constructed so as to insure acceptance by a common carrier or air express for safe transportation to the point of destination. Containers shall conform to the requirements of the Consolidated Freight Classification rules in effect at the time of shipment, except that fiberboard, when used, shall have a minimum Mullen Test of 275 pounds. Containers shall be able to withstand storage, rehandling, and reshipment without the necessity of repacking.

5.2 Marking. Marking shall be in accordance with MIL-P-10025.

5.3 Data cards. Data card information is not required.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number, and date of this specification.

6.2 Supersession data. This specification includes the requirements of Purchase Description PA-PD-1791, dated 25 May 1965.

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project No. 1195-A061

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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